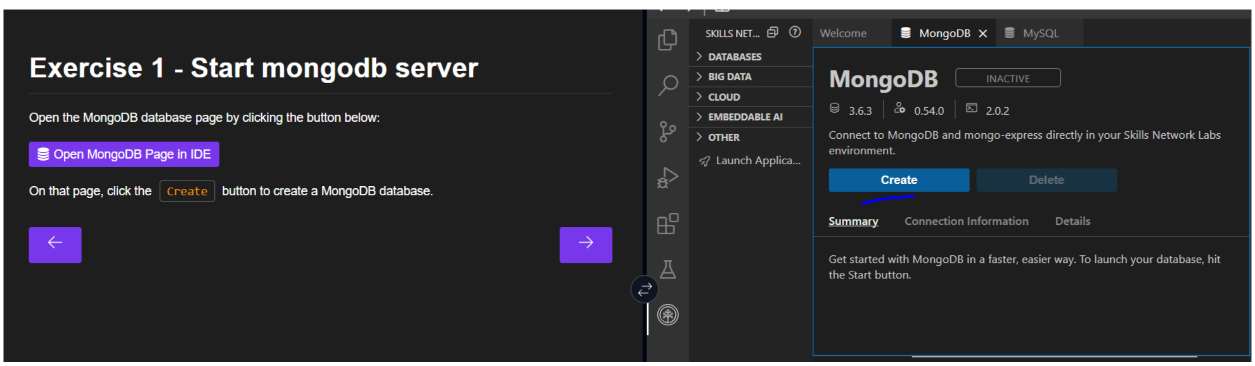
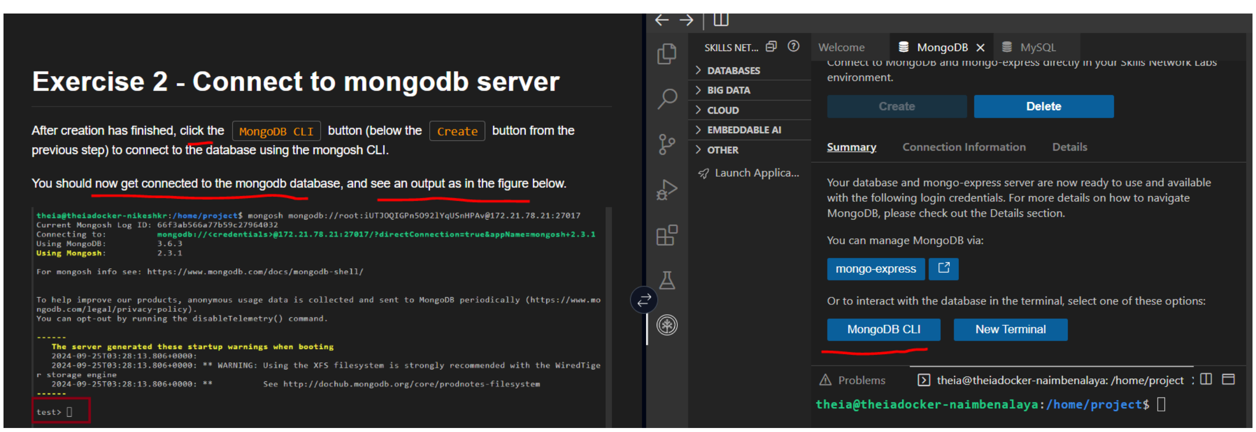
Getting Started with MongoDB

Objectives

After completing this lab you will be able to:

- Access the MongoDB server using the command-line interface
- Describe the process of listing and creating collections, which contain documents, and databases, which contain one or more collections
- Perform basic operations on a collection such as inserting, counting and listing documents





Exercise 3 - Find the version of the server

On the mongo client run the below command.

```
db.version()
```

This will show the version of the mongodb server.

```
theia@theiadocker-naimbenalaya:/home/project$ mongosh mongodb://root:Mz
18982Zdvl3ulASLl2S3DFf@172.21.94.229:27017
Current Mongosh Log ID: 686ea873677ceb71126b140a
Connecting to:          mongodb://<credentials>@172.21.94.229:27017/?di
rectConnection=true&appName=mongosh+2.4.2
Using MongoDB:          4.4.29
Using Mongosh:          2.4.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/


To help improve our products, anonymous usage data is collected and sen
t to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy
).
You can opt-out by running the disableTelemetry() command.

------
   The server generated these startup warnings when booting
   2025-07-09T15:39:08.075+00:00: Using the XFS filesystem is strongly
recommended with the WiredTiger storage engine. See http://dochub.mongo
db.org/core/prodnotes-filesystem
------

test> db.version()
4.4.29
test>
```

Exercise 4 - List databases

On the mongo client run the below command.

```
show dbs
```

This will print a list of the databases present on the server.

```
test> show dbs
admin    100.00 KiB
config    12.00 KiB
local     72.00 KiB
test>
```

Exercise 5 - Create database

On the mongo client run the below command.

```
use training
```

This will create a new database named **training**. If a database named training already exists, it will start using it.

Exercise 6 - Create collecton

On the mongo client run the below command.

```
db.createCollection("mycollection")
```

This will create a collection name **mycollection** inside the **training** database.

Exercise 7 - List collections

On the mongo client run the below command.

```
show collections
```

This will print the list of collections in your current database.

Exercise 8 - Insert documents into a collection

On the mongo client run the below command.

```
db.mycollection.insert({"color":"white","example":"milk"})
```

The above command inserts the json document {“color”:”white”,”example”:”milk”} into the collection.

Let us insert one more document.

```
db.mycollection.insert({"color":"blue","example":"sky"})
```

The above command inserts the json document {“color”:”blue”,”example”:”sky”} into the collection.

Insert 3 more documents of your choice.

Exercise 9 - Count the number of documents in a collection

On the mongo client run the below command.

```
db.mycollection.countDocuments()
```

This command gives you the number of documents in the collection.

Exercise 10 - List all documents in a collection

On the mongo client run the below command.

```
db.mycollection.find()
```

This command lists all the documents in the collection **mycollection**

Notice that mongodb automatically adds an '_id' field to every document in order to uniquely identify the document.

```
training> db.mycollection.insert({"color":"white","example":"milk"})
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('686ea952677ceb71126b140b') }
}
training> db.mycollection.insert({"color":"blue","example":"sky"})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('686ea957677ceb71126b140c') }
}
training> db.mycollection.countDocuments()
2
training> db.mycollection.find()
[
  {
    _id: ObjectId('686ea952677ceb71126b140b'),
    color: 'white',
    example: 'milk'
  },
  {
    _id: ObjectId('686ea957677ceb71126b140c'),
    color: 'blue',
    example: 'sky'
  }
]
training> 
```

Exercise 11 - Disconnect from mongodb server

On the mongo client run the below command.

```
exit
```

Practice exercises

1. Problem:

> *List databases.*

▼ Click here for Hint

> *Use the 'show' command with dbs option.*

▼ Click here for Solution

```
show dbs
```

2. Problem:

> *Create a database named **mydatabase**.*

▼ Click here for Hint

> *Use the 'use' command with the database name.*

▼ Click here for Solution

```
use mydatabase
```

3. Problem:

> *Create a collection named **landmarks** in the database **mydatabase**.*

▼ Click here for Hint

> *Use the 'createCollection' command.*

▼ Click here for Solution

```
db.createCollection("landmarks")
```

4. Problem:

> *List collections*

▼ Click here for Hint

> *Use the 'show' command with collections_ option.*

▼ Click here for Solution

```
show collections
```

5. Problem:

> *Insert details of five landmarks including name, city, and country. Example: Eiffel Tower, Paris, France.*

▼ Click here for Hint

> *Use the 'db.collection.insert()' command with the correct options.*

▼ Click here for Solution

```
db.landmarks.insert({"name":"Statue of Liberty","city":"New York","c
db.landmarks.insert({"name":"Big Ben","city":"London","country":"UK"})
db.landmarks.insert({"name":"Taj Mahal","city":"Agra","country":"India"})
db.landmarks.insert({"name":"Pyramids","country":"Egypt"})
db.landmarks.insert({"name":"Great Wall of China","country":"China"})
```

6. Problem:

> *Count the number of documents you have inserted.*

▼ Click here for Hint

> *Use the 'count' command on your collections.*

▼ Click here for Solution

```
db.landmarks.countDocuments()
```

7. Problem:

> *List the documents.*

▼ Click here for Hint

> *Use the 'db.collection.find()' command.*

▼ Click here for Solution

```
db.landmarks.find()
```

8. Problem:

> *Disconnect from the server.*

▼ Click here for Hint

> *Use the 'exit' command.*

▼ Click here for Solution

```
exit
```